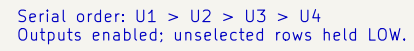
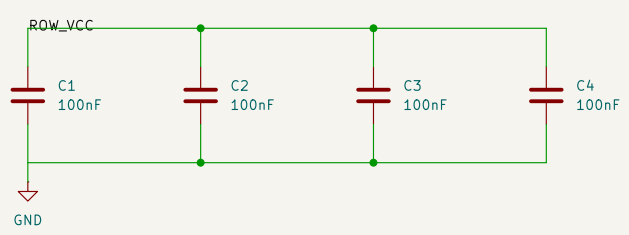


MCU serial stream > four latched drivers > 32-way row FFC

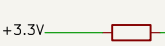


RDW_CLK_MCU.....

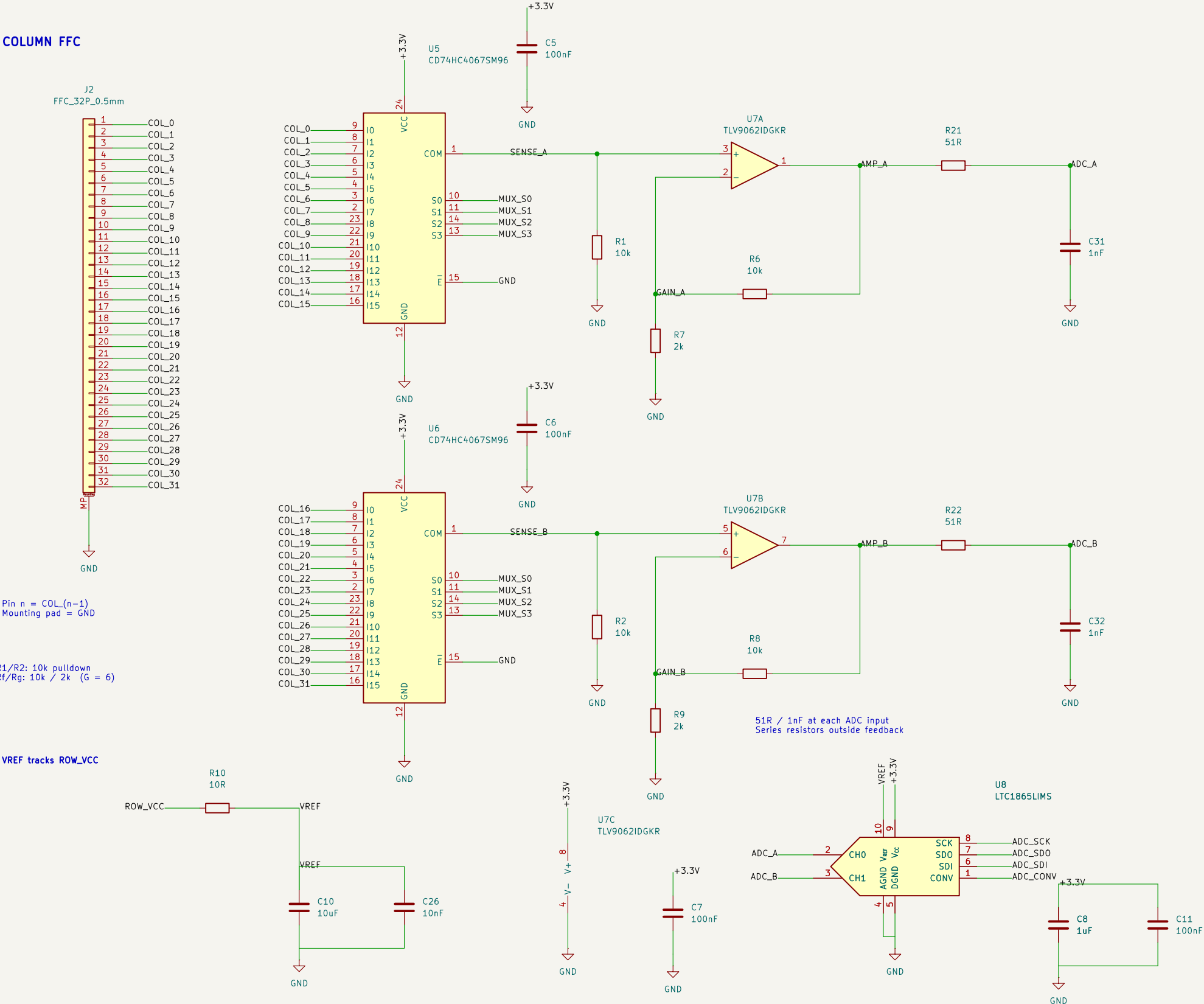
Clock / latch damping



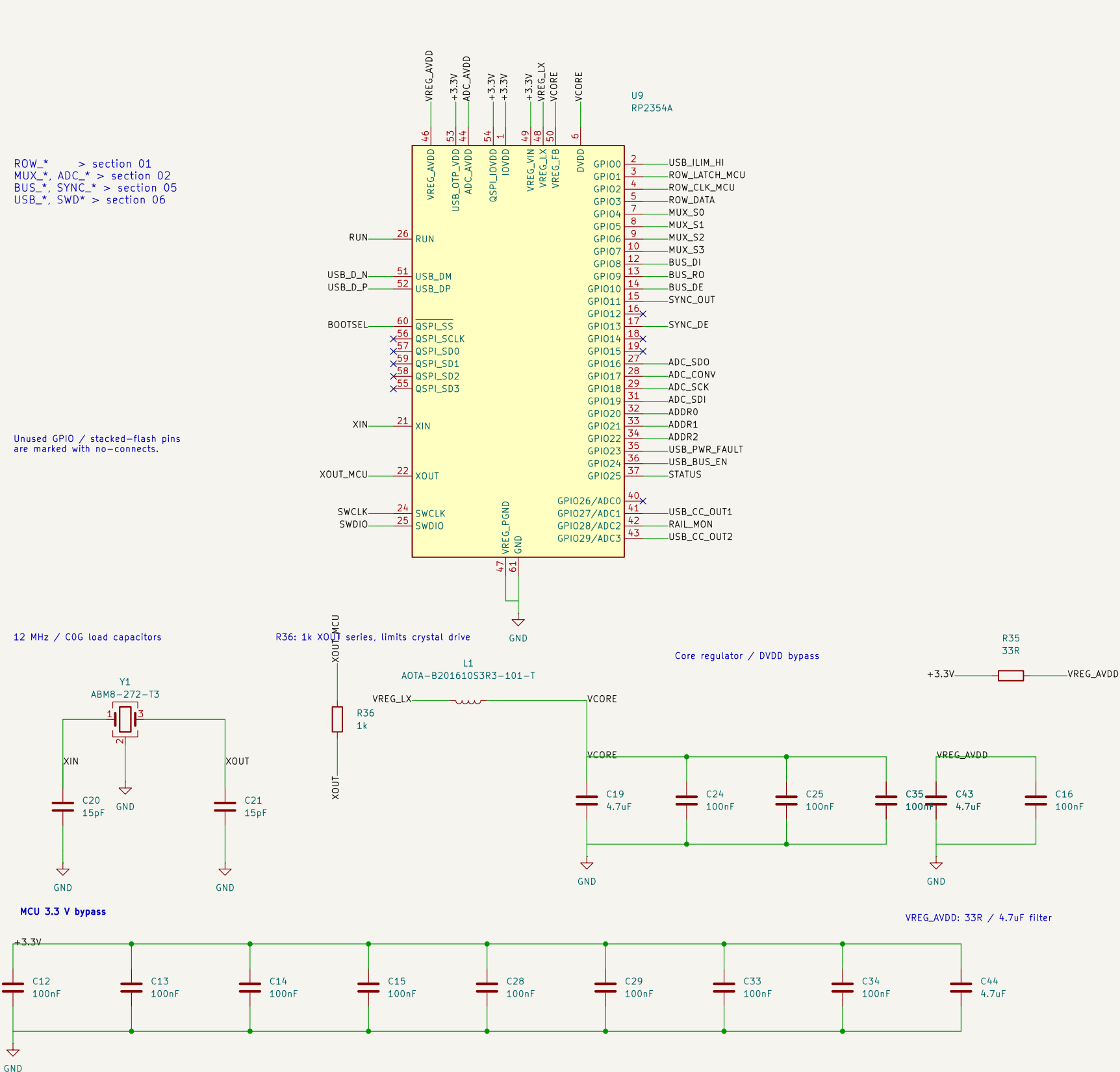
Excitation rail link



32 columns > two 16:1 muxes > gain of 6 > filtered ADC inputs



RP2354A / clock / core supply / local bypass



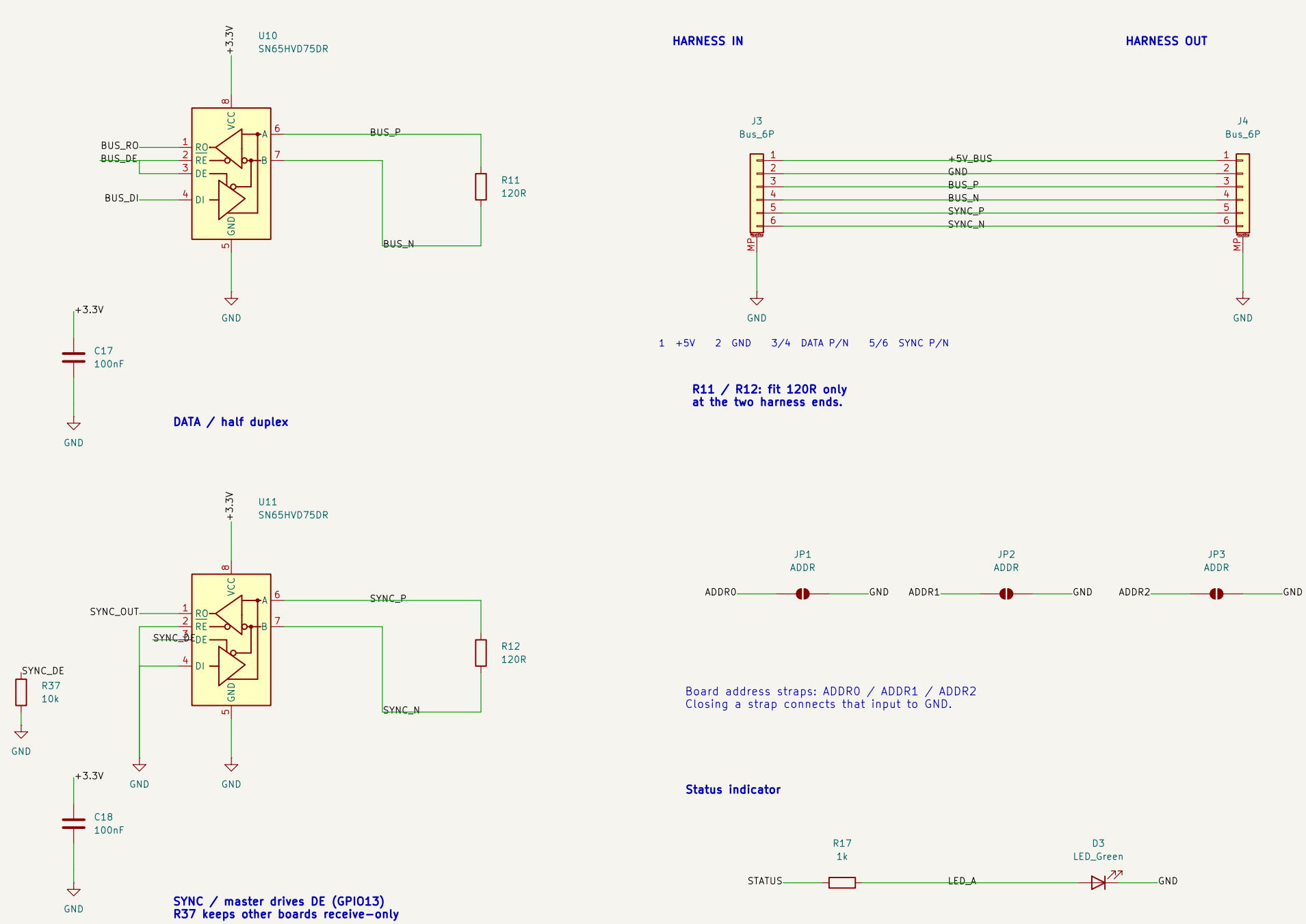
Harness or USB power > diode OR > local 3.3 V buck



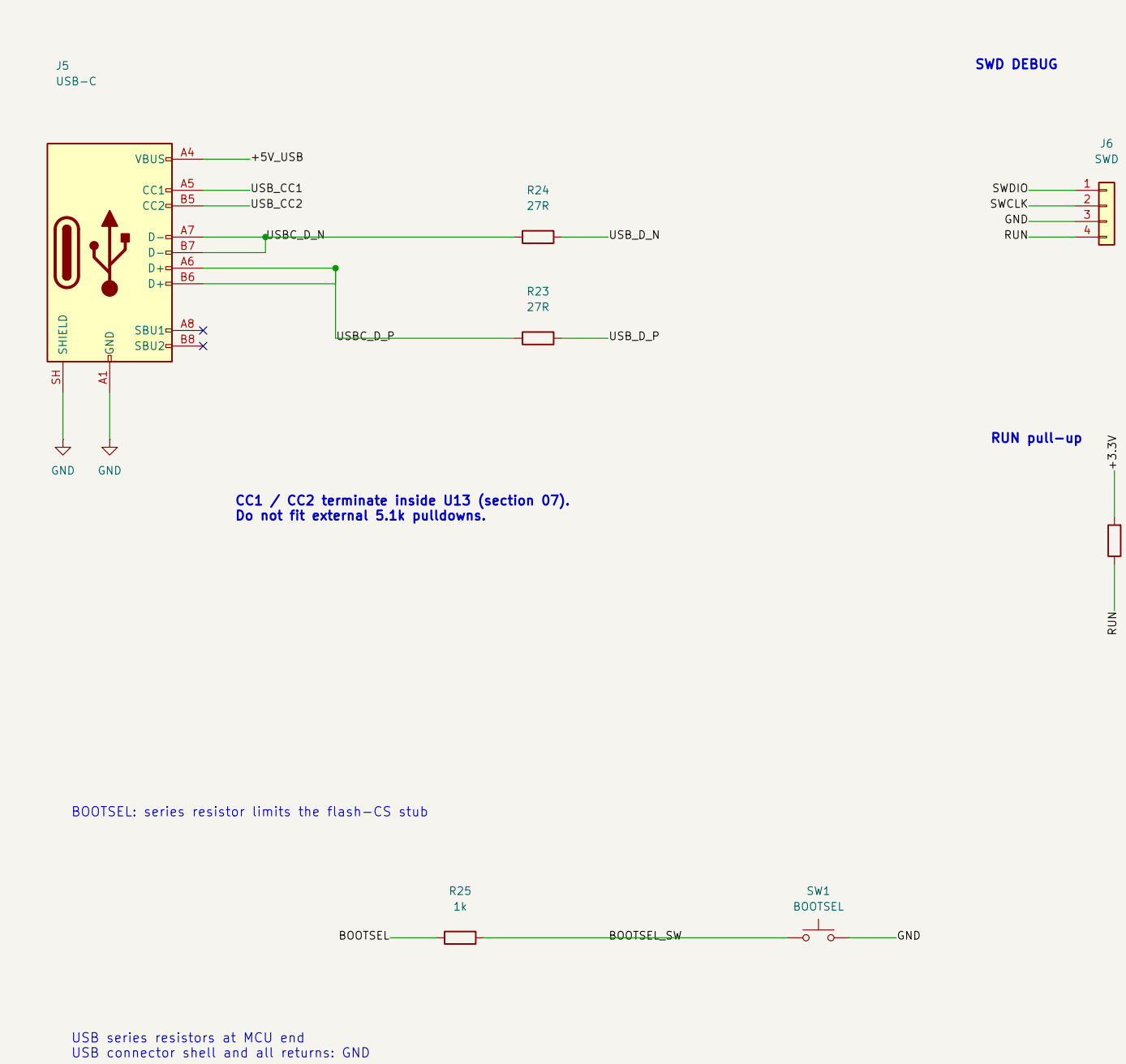
MCU analog supply filter

ROW_VCC link / VREF filter:
see sections 01 and 02.
All returns use GND.

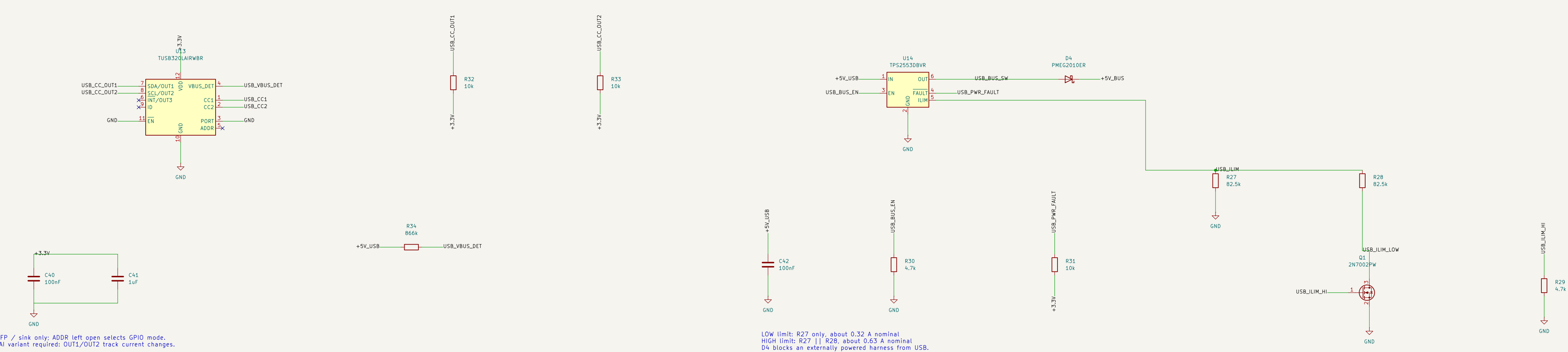
Half-duplex data pair and dedicated receive-only sync pair



USB-C data / configuration straps / SWD / BOOTSEL



Computer port current detection > firmware authorization > current limit > reverse barrier > +5V_BUS



FP / sink only: ADDR left open selects GPIO mode.
All variant required: OUT1/OUT2 track current changes.

LOW limit: R27 only, about 0.32 A nominal
HIGH limit: R27 || R28, about 0.63 A nominal
D4 blocks an externally powered harness from USB.

Reading the sheet

Matching net labels connect between sections. A dot marks a branch; an X marks an unused pin. Bypass capacitors are grouped by supply. USB attach current and the complete eight-board startup / operating budget require measurements; schematic checks cannot establish them.

3.3 V logic | USB-C data and protected chain power | SMD debug
Row drive / dual-bank memory / BR2354A / RS-485

Sheet: /

Title: TaxelScan | 32 x 32 sensor interface

KiCad E.D.A. 10.0.5	
15	

Id: 1/1